

Jason Lu

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EDUCATION

North Carolina State University, Raleigh, NC

Aug. 2022 – Dec. 2025

Bachelor of Science in Computer Science | Activities: AI/ML Teaching Assistant, HackPack

CERTIFICATIONS & SKILLS

AI/ML & Data Engineering: PyTorch, TensorFlow, NumPy, Pandas, SciPy, Jupyter Notebooks

Risk, Security & DevOps: Veracode, SonarQube, Azure, Azure DevOps, Docker, Jenkins, Linux, Bash

Backend & Data Architecture: Python, SQL, PostgreSQL, MySQL, Java, C/C++, Rust, Spring Boot, Hibernate, JUnit

Data Visualization & Automation: Streamlit, Plotly, Altair, Matplotlib, Seaborn, Selenium

Developer Tools & Frameworks: Git, GitHub, GitHub Copilot, Postman, React.js, Django, Maven, Makefiles

Other Technical Proficiencies: JavaScript, Kotlin, HTML/CSS, Figma, x86 Assembly

WORK EXPERIENCE

Junior Platform Engineer (MLOps), MetLife, Raleigh, NC

Feb. 2026 – Current

- Engineered an automated **Python** data pipeline to ingest and process unstructured text from **IBM OpenPages**, implementing advanced NLP preprocessing (tokenization, regex parsing, and text normalization) to convert raw regulatory data into structured feature vectors for RCM classification.
- Trained and evaluated Hugging Face models to generate accurate semantic embeddings, operationalizing the **Python** data pipeline as an **Azure Machine Learning (AML) Studio** endpoint to stream contextual data to an OpenAI model.
- Architected and deployed scalable **Azure** cloud infrastructure using **Terraform IaC**, contributing to internal open-source development models (**InnerSource**) to deliver reusable infrastructure modules across collaborative engineering teams.
- Designed and optimized robust **CI/CD** pipelines in **Azure DevOps**, integrating automated **Veracode** and **SonarQube** static analysis to enforce strict security guardrails and enhance overall code quality across platform components.

AI/ML Research Assistant, NCSU, Raleigh, NC

Jan. 2025 – Jan. 2026

- Executed comprehensive ablation studies across **XGBoost**, **LightGBM**, **LSTM**, and **Transformer** architectures, validating the stability of differentiable surrogate loss functions for time-series forecasting.
- Generated critical experimental results demonstrating a 33–50% reduction in Peak Mean Absolute Error (MAE) at 48-hour horizons, providing the empirical basis for the paper's core claims on optimization stability.
- Quantified trade-offs between predictive accuracy and operational safety via sensitivity analysis on asymmetric risk parameters, mitigating underprediction bias in critical risk-forecasting models.
- Designed a modular **Python**-based evaluation framework to automate hyperparameter tuning and model versioning, ensuring 100% reproducibility across distributed training runs on NCSU's high-performance computing (HPC) cluster.

Full Stack Development Intern, MyMatR, Raleigh, NC

Aug. 2025 - Dec. 2025

- Optimized full-stack performance by systematically troubleshooting integration bottlenecks between **React** frontends and **Django** backends, reducing page load latency by 3 seconds.
- Architected a scalable **PostgreSQL** database and **Django** REST API ecosystem, enforcing data integrity through advanced ORM constraints, validations, and atomic transactions to streamline data flows.
- Formulated a comprehensive test automation framework by authoring backend unit tests with **pytest** and frontend integration tests for **React** components, establishing rigorous quality gates to prevent regression defects.
- Utilized **Git** and **GitHub** for version control, actively participating in team code reviews, pull request workflows, and continuous integration practices to maintain codebase stability.

Full Stack Engineer Intern, MetLife, Cary, NC

Jun. 2025 – Aug. 2025

- Engineered features for a policy search portal using **Azure AI Search**, implementing semantic search capabilities that improved result relevance ranking by 15% compared to keyword-based baselines.
- Developed an automated web scraping solution using **Python** and **Selenium** to navigate complex government portals, reliably extracting and downloading critical regulatory documents for internal processing.
- Deployed and managed search application components on **Microsoft Azure**, gaining hands-on experience with cloud-native data services and scalable architecture patterns.
- Collaborated within an **Agile Scrum** team to refine product requirements, translating user needs into technical specifications for data-driven features.